Quantum Mechanical Calculations of Metal Ion Intercalation in Bilayer Graphene for Neuromorphic Applications

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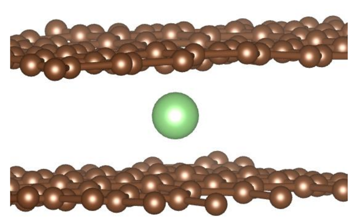
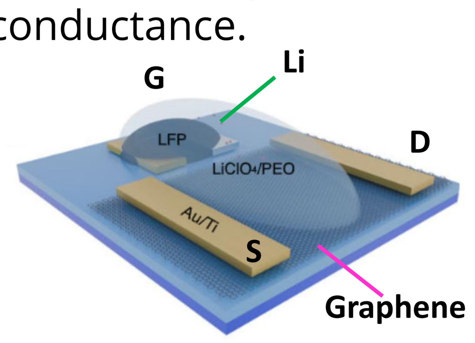
Background

Neuromorphic Computing: Classic computer architecture has a limited throughput between memory and processing. Inspired by synapses in the human brain, neuromorphic architectures use in-memory computation through artificial synapses that store data as analog conductance.

Scan for Neuromorphic Computing overview



Bilayer Graphene with Li



ECRAM

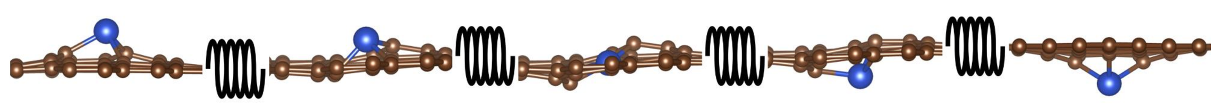
Bilayer graphene ECRAM Synapses: Electrochemical random-access memory (ECRAM) uses metal ion movement through a channel to program conductance. With this device, lithium intercalates between graphene layers, changing carrier density and therefore tuning conductance.

Research Goal: Use ab-initio calculations to characterize lithium intercalation, providing a theoretical basis for designing graphene-based synapses.

Methodology

DFT: Density Functional Theory (DFT) uses the many-body Schrodinger Equation and Kohn-Sham equations to simplify the solving process of multi-particle systems using electron density instead of individual wavefunctions.

NEB: Nudged Elastic Band method finds the minimum energy pathway between a starting and ending configuration by treating intermediary images like they're connected with springs.



AIMD: Molecular dynamics (MD) studies how a molecular system moves over time. Ab initio MD uses DFT to estimate the forces acting on each atom and find their position after a time step using Newton's Law.

Post-Analysis: After using Vienna Ab initio Simulation Package (VASP) to apply the above techniques, calculations will be carried out to further characterize ion intercalation.

Bader Charge Analysis: Define atom boundaries along the points in space where charge density is at a minimum. Find the charge on an atom by integrating the charge density with respect to volume within the atom's boundaries.

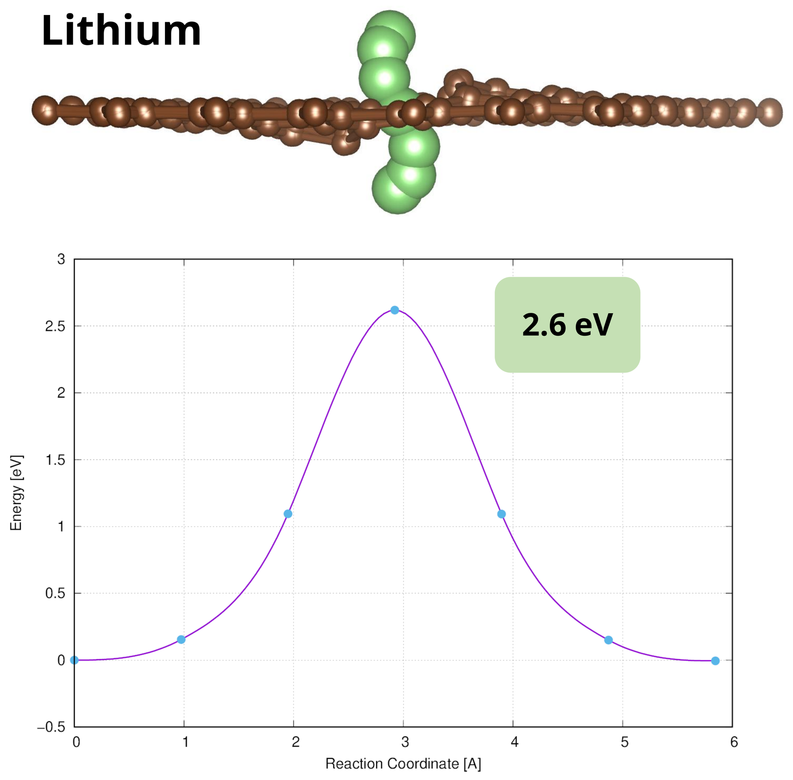
TAMSD: Measure how lithium diffuses inside bilayer graphene using Time-Averaged Mean Squared Displacement (TAMSD) of the lithium's position over time.

$$TAMSD(t) = \frac{1}{T-t}\sum_{n=0}^{T-t-1}[r(t+n) - r(n)]^2$$

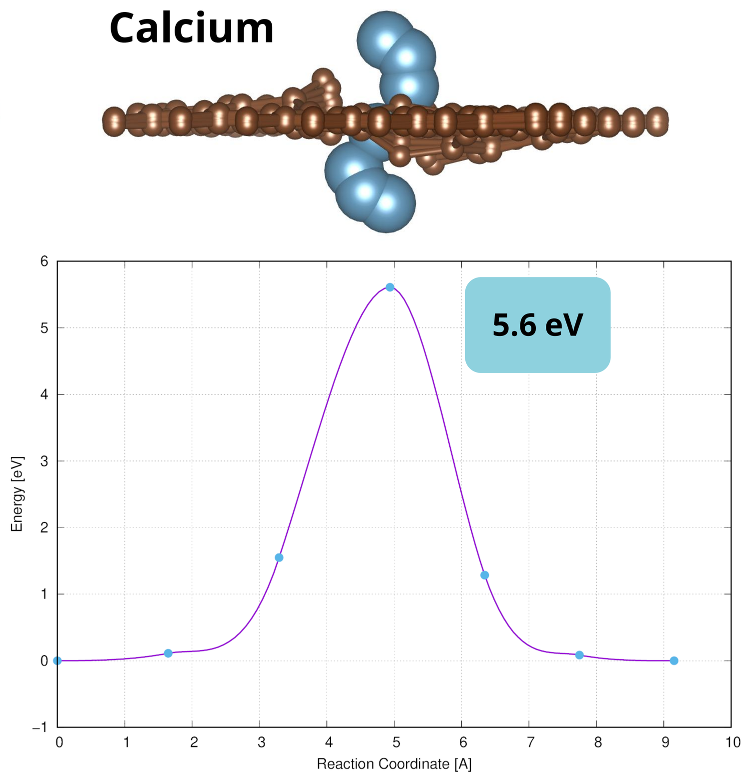
T = # time steps
$r_i(t)$ = position of atom at time t

Results & Analysis

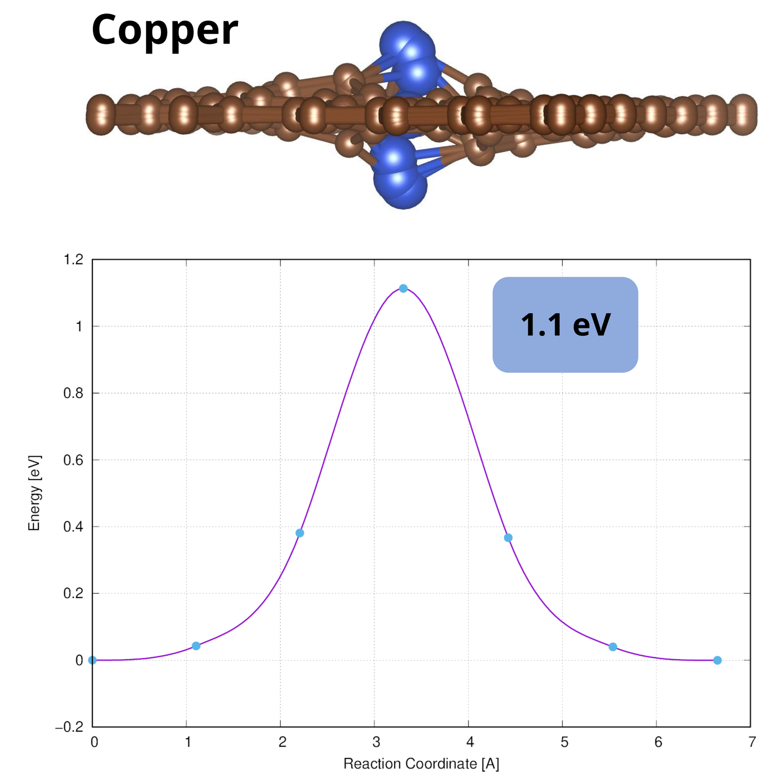
Lithium



Calcium

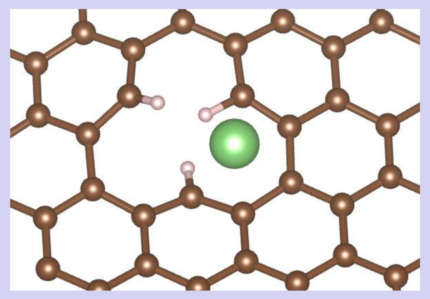


Copper



Hydrogen Bond Termination:

Energy Barrier: 1.74 eV (compared to 2.6 eV)

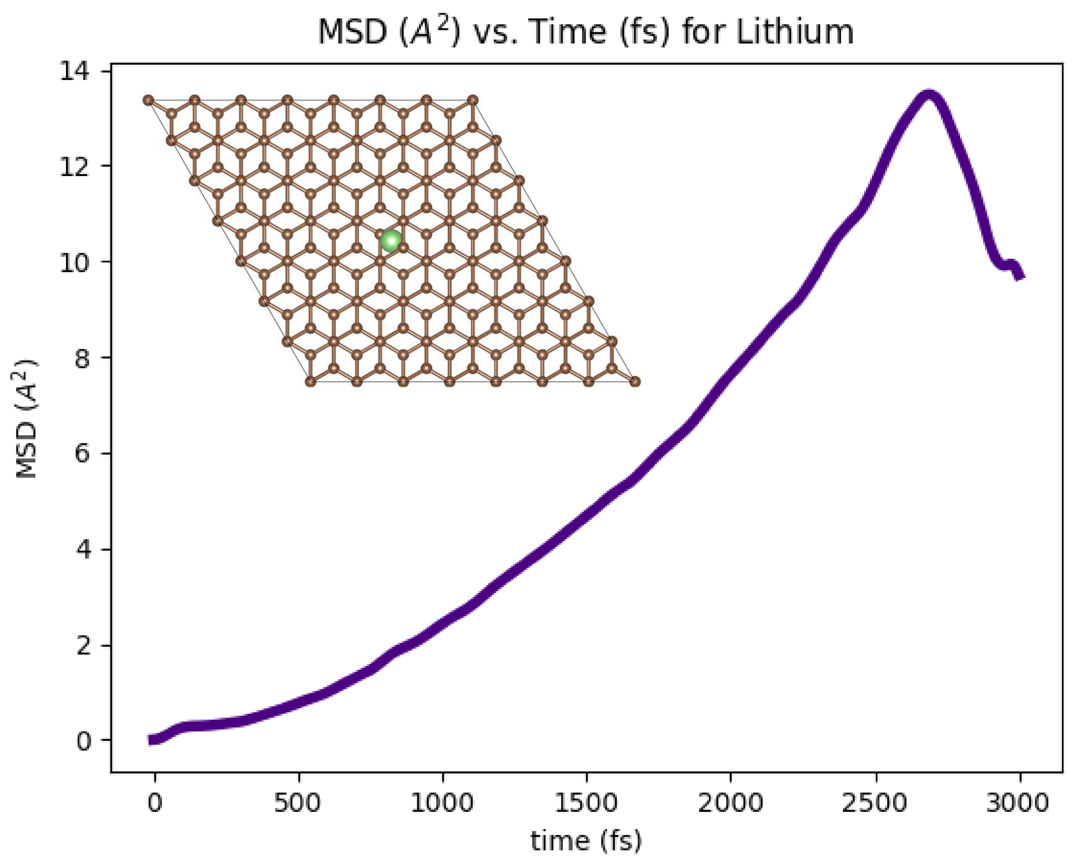


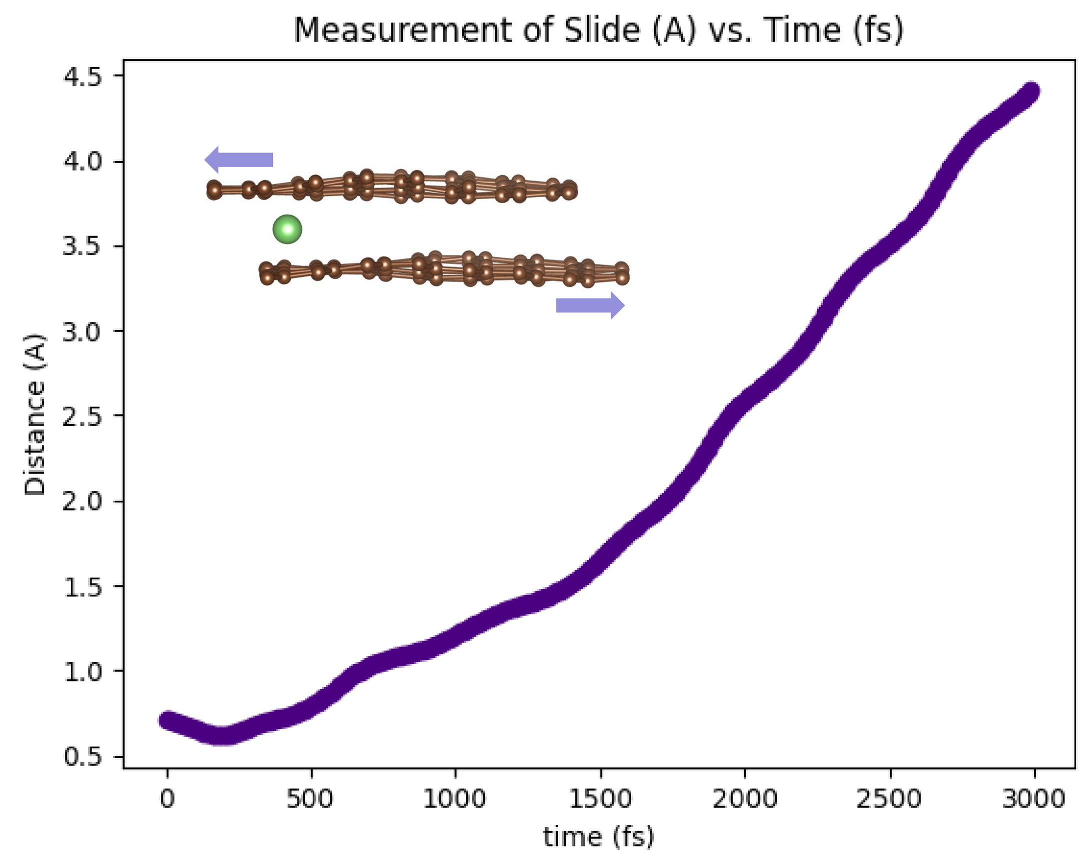
Metal Ion:

- **Li+:** 1.54 eV Energy Barrier
- **Ca2+:** 6.67 eV Energy Barrier
- **Mg2+:** 3.15 eV Energy Barrier

Bader Charge Analysis:

- **Li+:** +0.893 Charge on Lithium
- **Ca2+:** +1.488 Charge on Calcium





Diffusion Coefficient using AIMD:

Variables:

- Supercell: 7x7
- Temperature: 300K
- Time Step: 1 fs (10^{-15} s)
- Total Steps: 3000 (3 ps)
- Bottom Graphene Layer: Frozen

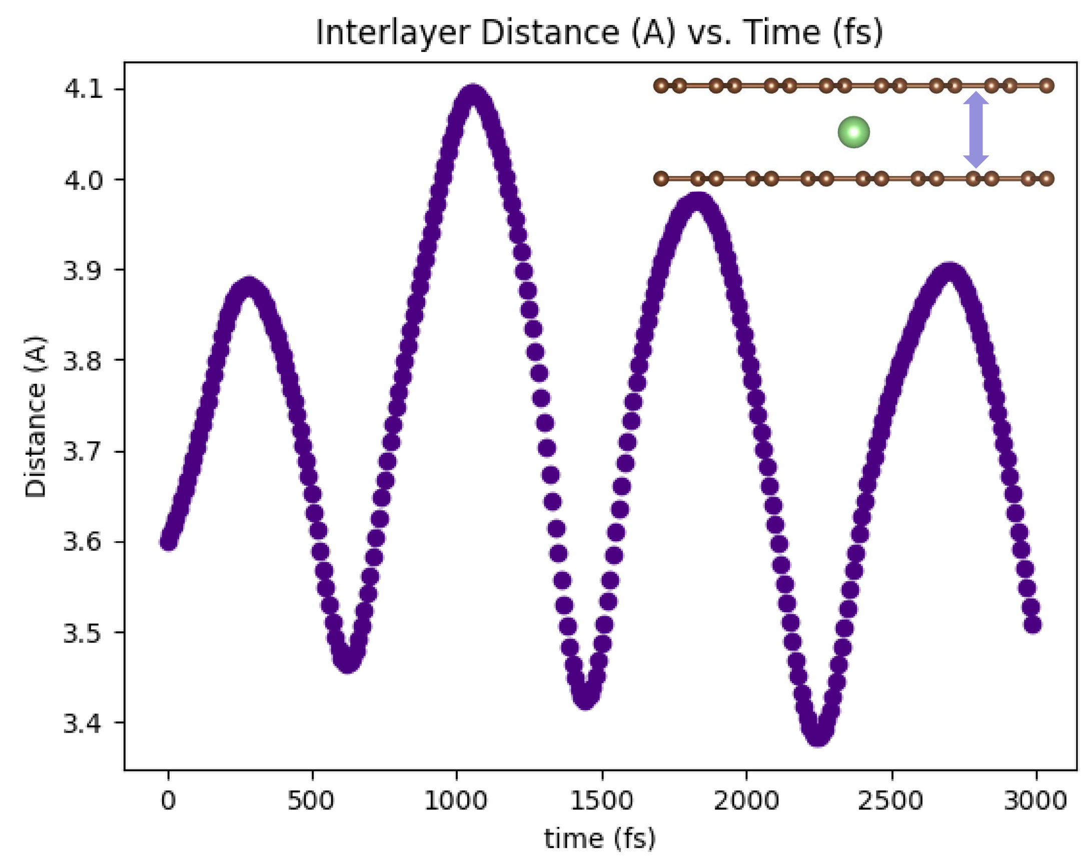
Calculation:

Einstein Equation and relation for diffusion:

$$D = \lim_{t\to\infty}\frac{MSD(t)}{6t} \rightarrow D = \frac{MSD'(t)}{6}$$

Result: $D = 5.717 \times 10^{-5}\ cm^2/s$

*Dependent variables in other trials: Supercell size (5x5), temperature (600K), time step (3 fs), bottom graphene layer (free), K-point sampling, pseudopotentials



Conclusions

Hydrogen bond termination decreases Li diffusion barrier through monovacancy by 0.9 eV

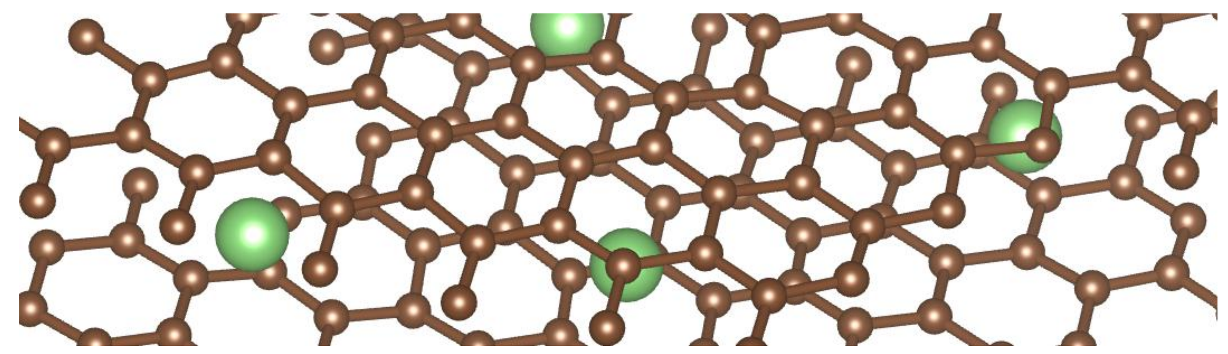
Li+ has a 1.06 eV lower diffusion barrier than Li, whereas Ca2+ has a 1.07 eV higher diffusion barrier than Ca and Mg2+ has a 0.41 eV higher diffusion barrier than Mg

Diffusion Coefficient of Li in bilayer graphene with AIMD and TAMSD:	Diffusion Coefficient from experimental results in literature:
$5.7 \times 10^{-5}\ cm^2 s^{-1}$	10^{-6} to $7 \times 10^{-5}\ cm^2 s^{-1}$

Interlayer distance of the graphene oscillates at 1.2THz by 0.07nm, while the sheets appear to slide apart linearly

Future Work

- 2-Carbon Vacancy metal ion intercalation
- Li+ diffusion in AA and AB bilayer graphene with NEB
- AIMD to explore complex diffusion and insertion mechanics such as multi-Li interaction and charged states



- Optimize experiment parameters and direction by comparing AIMD results to pure DFT results

Acknowledgements

- I would like to thank my mentors Simon Cao and Professor Anantram for their continued guidance and support
- This research is supported by the Mary Gates Endowment for Students and the NSF FuSe (2328815) Award
- I would like to acknowledge the Quantum Devices Lab for the access to VASP and the University of Washington's high-performance compute cluster Hyak-Klone for the computation resources

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